VISTULA UNIVERSITY

The Faculty of Computer Engineering, Graphics Design, and Architecture

Program of Study: Computer Engineering

Design and Comparative Evaluation of NLP Models and Storage Strategies for Large-Scale Steam Review Analytics

Bachelor's Thesis • Warsaw, 2026

Author: Nihal Singh (Student No. 77963)

Supervisor: Professor Dmitriy Kostiuk

Why This Study

Sarcastic, informal gaming language breaks traditional sentiment tools — and querying a million reviews efficiently is its own challenge

991,650

Steam reviews analyzed

242

games in the corpus

3

NLP models compared

Three Research Objectives

NLP Accuracy

Can a fine-tuned transformer (BERT) beat lexicon tools VADER & TextBlob on sarcastic reviews? → H1

Storage Performance

Benchmark PostgreSQL, SQLite, Parquet & CSV on text-heavy analytical queries. → H7, H6

Success Prediction

Which game-level features (playtime, sentiment) best predict commercial success? → H3, H2, H4

Formalised into 7 Hypotheses

- **H1** BERT F1 ≥ 90%, beats VADER/TextBlob by ≥20pp
- **H2** Positive early reviews → 3× owners at 6 months
- **H3** Playtime features > 30% SHAP importance
- **H4** XGBoost classifies success with > 85% accuracy
- **H5** LDA finds ≥ 8 coherent review topics
- **H6** API responds < 2s on 500K reviews
- **H7** PostgreSQL+GIN beats SQLite; Parquet compresses best

Methodology

Kaggle Steam Games Reviews dataset, a 10-step NLP pipeline, and a controlled 4-format storage benchmark

Dataset & NLP Pipeline

Source: Kaggle — Steam Games Reviews and Rankings

Scope: 991,650 reviews · 242 games (2010–2023)

Ground truth: is_recommended (82.2% / 17.8%)

Pipeline: 10-step clean → VADER · TextBlob · BERT

BERT: bert-base-uncased, 40K reviews, 3 epochs, Tesla T4

Storage Benchmark Setup

Formats: PostgreSQL 15 · SQLite · Parquet · CSV

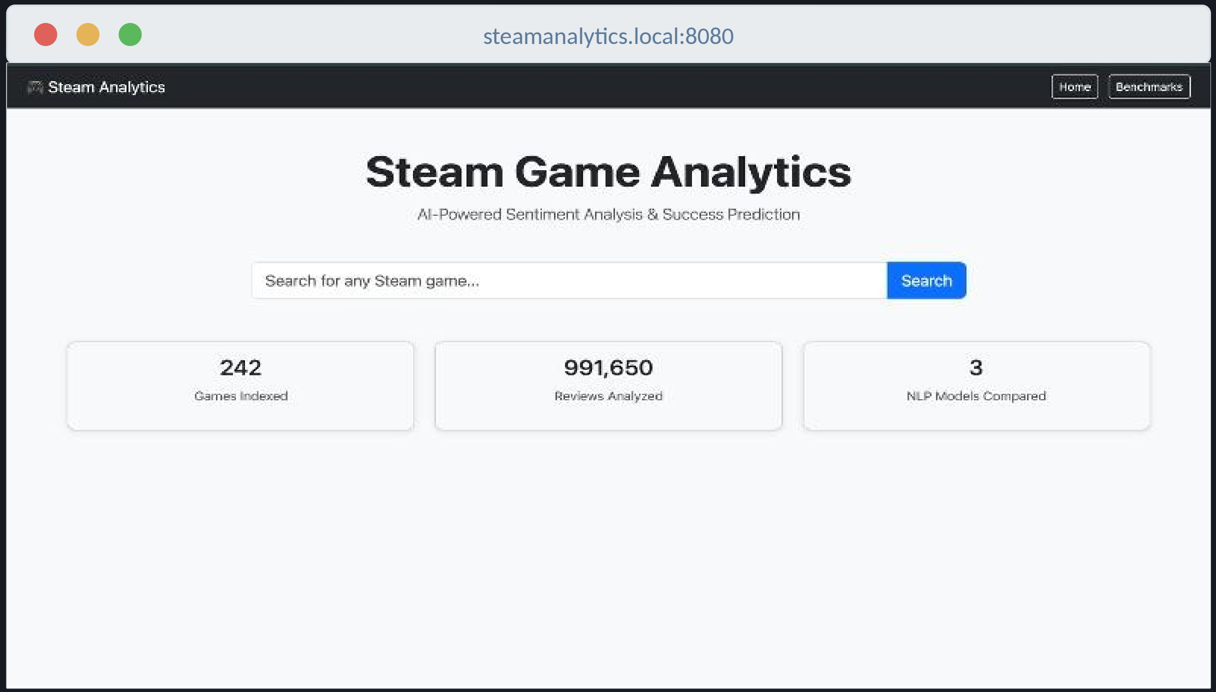
Workload: 9 query types, 5 reps, warm-up excluded

Indexes: B-tree, GIN, composite B-tree

Tested on a held-out 10,000-review set, identical for all 3 NLP models.

Deployed Platform

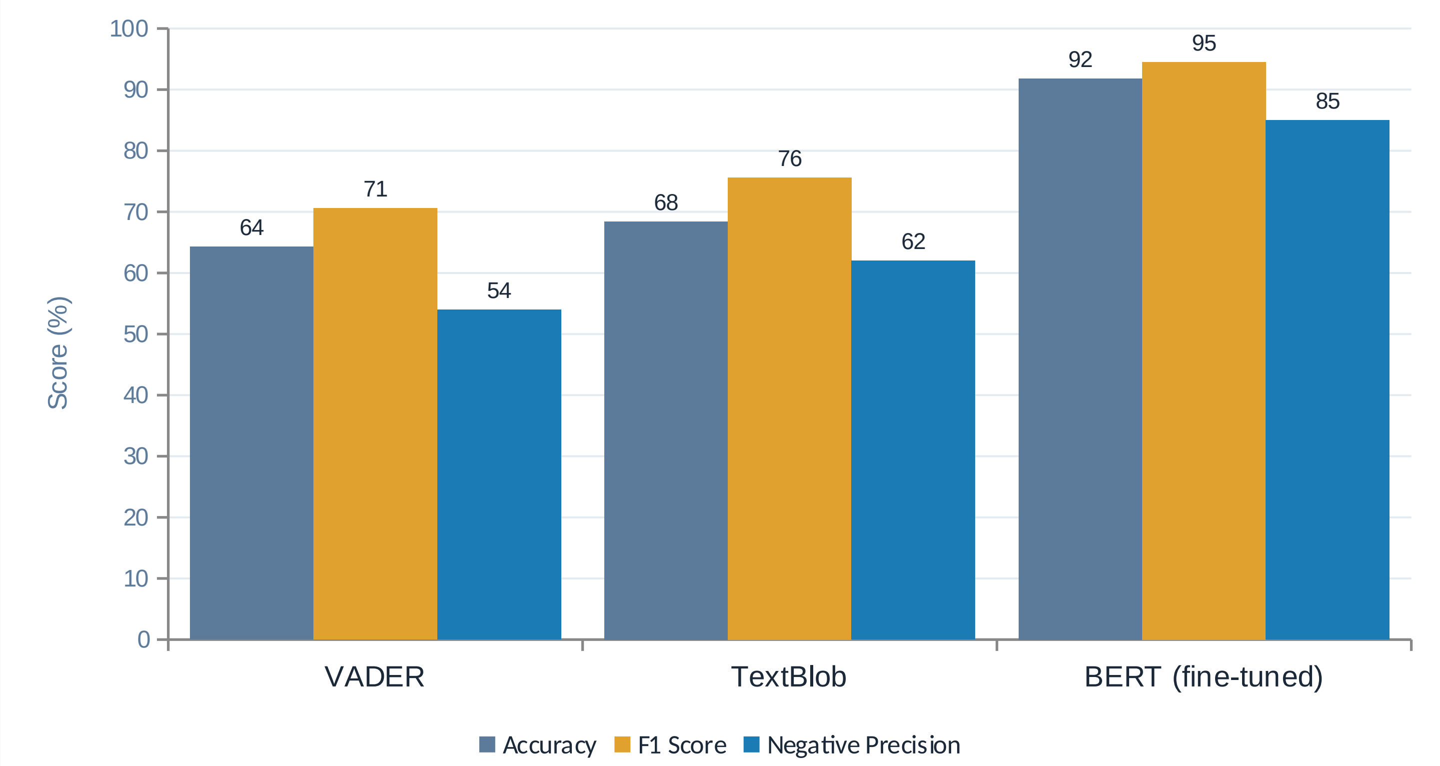
Live Flask + PostgreSQL dashboard — Steam Game Analytics (Fig. 5.4)



Search across 242 games · 991,650 reviews · 3 NLP models compared, served by 5 REST endpoints

NLP Model Comparison — H1

Fine-tuned BERT vs. lexicon-based VADER and TextBlob on 50,000 Steam reviews



94.5%

BERT F1 score — exceeds the 90% threshold and beats VADER by 23.9 points

Why VADER fails

Double negation: “not...isn't a good port” scored +0.99

Contextual tone: “not of hatred, but of love” scored +0.99

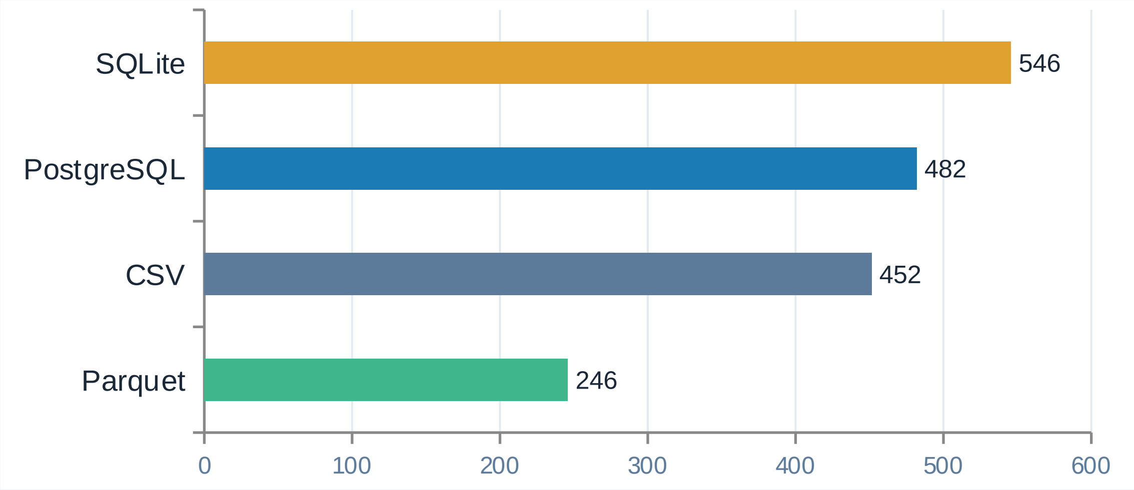
Temporal shifts: “great in 2017. Fell off.” scored positive

H1 CONFIRMED — BERT clears every threshold on every metric

Storage Benchmarking — H7

No single format wins everywhere — performance depends on the workload

Storage Size — 991,650 reviews



Parquet (Snappy) is 45.5% smaller than raw CSV; PostgreSQL & SQLite trade space for index speed.

Which format wins which query?

Query	Best Format	Time
Q1 Point lookup	SQLite	16.8 ms
Q3 Aggregation	Parquet	128.3 ms
Q5 Top-K query	Parquet	117.5 ms
Q6 Full scan	Parquet	38.9 ms
Q7 GIN text + filter	PostgreSQL	540.8 ms
Q9 Rare-word search	PostgreSQL	1,353.9 ms

73× faster full-scan with Parquet vs. CSV (38.9 ms vs 2,842.6 ms)

2.5× faster indexed text search: PostgreSQL+GIN vs. SQLite (Q7)

H7 CONFIRMED — workload, not a single "best" engine, determines the optimal storage format

Additional Findings

Topic modeling, behavioural success prediction, and live platform performance

Topic Modeling — H5

10

coherent topics found (c_v = 0.4767) across 43,342 reviews

All 4 expected themes surfaced:

bugs, story, gameplay & genre clusters

Plus: Early Access pattern & modding-community topic

CONFIRMED

Success Prediction — H3

33.5%

SHAP importance of pct_refund_window — the single strongest predictor

% of reviewers playing < 1 hour

outpredicts review sentiment text entirely.

All hours-based features: 33.0% SHAP (> 30% threshold)

CONFIRMED

Platform Performance — H6

252.7ms

worst-case Flask API response on 500,000 reviews — vs. a 2,000 ms threshold

Game Stats endpoint: 3.8 ms avg

Reviews endpoint: 7.0 ms avg

Search (ILIKE) endpoint: 251.0 ms avg

CONFIRMED

Conclusions & Hypothesis Verdicts

5 of 7 hypotheses confirmed; H2 and H4 skipped for lack of data, not failure of the hypothesis

ID	Hypothesis	Result	Verdict
H1	BERT F1 ≥ 90%, beats VADER/TextBlob by ≥20pp	F1 = 94.5%	CONFIRMED
H2	Early positive reviews → 3× owners at 6mo	no ownership data	SKIPPED
H3	Playtime features > 30% SHAP importance	33.0%	CONFIRMED
H4	XGBoost classifies success > 85% accuracy	242 games too few	SKIPPED
H5	LDA finds ≥ 8 coherent topics	10 topics, c_v 0.4767	CONFIRMED
H6	API responds < 2s on 500K reviews	252.7 ms worst case	CONFIRMED
H7	PostgreSQL+GIN beats SQLite; Parquet compresses	all sub-claims hold	CONFIRMED

Key Contributions

- Fine-tuned BERT (94.5% F1) decisively outperforms lexicon-based sentiment tools on sarcastic gaming text
- First study to benchmark PostgreSQL+GIN against SQLite and Parquet on a single ~1M-row text corpus
- Early player drop-off (pct_refund_window) is a stronger success signal than review sentiment itself